

JASON POULOS

Research Scientist · AI Evaluations & Agents · Machine Learning

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SUMMARY

Research scientist at the intersection of machine learning, causal inference, and policy, focused on evaluating frontier AI by designing frameworks that translate safety concerns and model capabilities into measurable tests. Co-author of **Terminal-Bench** (ICLR 2026) and **Humanity's Last Exam** (*Nature* 2026). Previously built interpretable transformers on 10M+ clinical events at Mass General Brigham and novel causal inference methods for drug safety at Harvard Medical School; award-winning work on Bayesian adversarial ML (JASA).

OPEN-SOURCE & BENCHMARK PROJECTS

Terminal-Bench: Benchmarking AI Agents in Command-Line Interfaces

2025 – 2026

Co-author and open-source contributor; Stanford University & Laude Institute

- Co-authored **Terminal-Bench** (ICLR 2026), a benchmark of hard, realistic tasks for frontier AI agents in command-line interfaces; merged 17 pull requests, including 8 original benchmark tasks.
- Engineered scientific-computing environments that translate highly specialized real-world problems into autonomous RL environments with custom graders, systematically identifying failure modes in long-horizon agent execution.
- Built the *parallelize-graph* task, an evaluation at the intersection of high-performance computing and bioinformatics that requires agents to implement parallel de Bruijn graph construction for genome assembly in Unified Parallel C (UPC), managing intricate build systems and multi-step verification in the terminal (**featured in the Terminal-Bench task spotlight**).
- Designed further tasks on fMRI encoding models, stable parallel k -means, parallel particle simulation, adaptive rejection sampling, the cross-entropy method, and attention-based multiple-instance learning.
- Improved infrastructure: a task-difficulty rubric, a LiteLLM fix for the Terminus agent, and post-release grader fixes.

Humanity's Last Exam: A Benchmark of Frontier AI Capabilities

2025

*Contributed expert-level questions; credited co-author of the *Nature* (2026) paper; Center for AI Safety & Scale AI*

EXPERIENCE

Human Frontier Collective Specialist (Contract)

2025

Scale AI

Remote

- Designed evaluation frameworks and domain-specific problem sets that test frontier models across diverse capabilities.
- Trained machine-learning-engineer agents by creating expert solutions to machine learning benchmark competitions.

Postdoctoral Research Fellow in Machine Learning

2023 – 2024

Brigham and Women's Hospital & Harvard Medical School (Mass General Brigham)

Boston, MA

- Developed interpretable transformer neural networks on large-scale Mass General Brigham electronic health records (over 10 million clinical events), combining static and temporal patient features to predict diabetes treatment failure ($\text{HbA1c} \geq 7\%$ within 12 months) and enable early, targeted interventions (**code**).
- Benchmarked tabular transformers (TabTransformer, FT-Transformer) against a GPT-style autoregressive transformer trained on patient event streams; built the end-to-end PyTorch pipeline for preprocessing, multi-GPU training, and Ray Tune hyperparameter search.
- Applied explainability methods to surface clinically meaningful risk drivers (e.g., HbA1c variability), working with endocrinologists to make model outputs interpretable and actionable for care teams.

Postdoctoral Fellow in Data Science

2021 – 2023

Department of Health Care Policy, Harvard Medical School

Boston, MA

- Extended Targeted Minimum Loss-Based Estimation (TMLE), a doubly robust causal ML method, to multi-valued treatments to evaluate the long-term safety of six antipsychotic drugs for patients with severe mental illness using large-scale claims data (lead author, *Statistics in Medicine* 2024; **code**).
- Showed in simulation that the estimator stays consistent under model misspecification, and quantified diabetes and mortality risk across diagnostic subgroups (*Psychological Medicine* 2023; *BJPsych Open* 2024).
- Extended the method to longitudinal treatment regimes by embedding recurrent neural networks, reducing bias relative to ensemble learners in simulations; released open-source **simulation code**.

- Postdoctoral Associate, Causal Inference and Deep Learning Programs** 2019 – 2021
Statistical and Applied Mathematical Sciences Institute (SAMSI) & Duke University Durham, NC
- Co-developed Bayesian frameworks for adversarial machine learning that relax the unrealistic common-knowledge assumptions of game-theoretic defenses, yielding more robust, reliable models (*JASA* 2023; Premio SEIO–Fundación BBVA 2026; [code](#)).
 - Benchmarked deep generative imputation (GAIN, denoising autoencoders) against tree-based multiple imputation on American Community Survey data: trees achieved lower bias and better coverage, while deep models ran faster (*Survey Methodology* 2022; [code](#)).
 - PI on NSF supercomputing allocations (TACC Frontera, XSEDE) to train RNNs on high-dimensional health data.

- Graduate Researcher (NSF Graduate Research Fellow)** 2013 – 2019
University of California, Berkeley Berkeley, CA
- Developed RNN encoder–decoder models for counterfactual time-series prediction with propensity-weighted losses (*JRSS-C* 2021; [code](#)), and matrix-completion methods for causal inference on panel data (*Statistics and Public Policy* 2024; [code](#)).
 - Built character-level attention encoder–decoder networks that transcribe handwritten text without segmentation or bounding boxes (*Neural Computing & Applications* 2021; [code](#)).
 - Generalized results from a randomized experiment with noncompliance to a target population using ML ensembles, applied to a large Medicaid insurance experiment (*Journal of Causal Inference* 2020; [code](#)).

TECHNICAL SKILLS

AI evaluation: agent benchmarks, RL environments and automated graders, LLM capability evaluations, failure-mode analysis, containerized task harnesses (Terminal-Bench, Harbor)

Machine learning: transformers (tabular and temporal), RNN and encoder–decoder models, attention, GANs and autoencoders, adversarial robustness, explainability

Statistics & causal inference: experiments with noncompliance, doubly robust estimation (TMLE), propensity scores, matrix completion, Bayesian modeling, multiple imputation

Languages: Python, R, SQL, Bash, C/C++, UPC

Tools: PyTorch, TensorFlow/Keras, scikit-learn, Ray Tune, LiteLLM, Docker, Git/GitHub, Linux, CUDA multi-GPU training, Open MPI, HPC (TACC Frontera, XSEDE)

EDUCATION

- Ph.D., Political Science, Designated Emphasis in Computational Science & Engineering, UC Berkeley** 2019
M.A., Political Science, UC Berkeley 2014
Graduate Special Student (non-degree coursework in Economics), *Massachusetts Institute of Technology* 2011–2013
Non-degree graduate coursework in Economics, Harvard University 2010
B.A., Economics, Phi Beta Kappa, University of Massachusetts Amherst 2008

SELECTED PUBLICATIONS

- M. A. Merrill, A. G. Shaw, N. Carlini, ..., **J. Poulos**, ..., L. Schmidt (2026). “Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces.” *International Conference on Learning Representations (ICLR)*. [[arXiv](#)] [[project](#)] [[blog](#)]
- Center for AI Safety, Scale AI, & HLE Contributors Consortium (2026). “A benchmark of expert-level academic questions to assess AI capabilities” [Humanity’s Last Exam]. *Nature*, 649: 1139–1146. [[arXiv](#)] [[project](#)]
- D. Rios Insua, R. Naveiro, V. Gallego, and **J. Poulos** (2023). “Adversarial Machine Learning: Bayesian Perspectives.” *Journal of the American Statistical Association*, 118(543): 2195–2206. Premio SEIO–Fundación BBVA 2026. [[arXiv](#)] [[code](#)]
- J. Poulos**, M. Horvitz-Lennon, K. Zelevinsky, et al. (2024). “Targeted Learning in Observational Studies with Multi-Valued Treatments: An Evaluation of Antipsychotic Drug Treatment Safety.” *Statistics in Medicine*, 43(8): 1489–1508. [[arXiv](#)] [[code](#)]
- J. Poulos** and S. Zeng (2021). “RNN-Based Counterfactual Prediction, with an Application to Homestead Policy and Public Schooling.” *Journal of the Royal Statistical Society, Series C*, 70(4): 1124–1139. [[arXiv](#)] [[code](#)]

HONORS, SERVICE & TALKS

- Honors:** Premio SEIO–Fundación BBVA (2026); NSF Graduate Research Fellowship (2013–2018)
- Reviewer:** NeurIPS, ICLR, ICML, AAAI, AISTATS, UAI, ML4H, MLHC; NeurIPS ethics review (2023–2025); 15+ journals
- Invited talks:** Carnegie Mellon, University of Toronto, Brandeis, Banff International Research Station; 20 conference talks